

Artificial Intelligence & Machine Learning — Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

California Housing Dataset

1. Overfitting Analysis

A model overfits when it performs very well on training data but poorly on new, unseen data. This happens when the model learns noise in the training set rather than actual patterns. The most reliable indicator is a large difference between training error and test error.

Baseline: Unconstrained Decision Tree

A Decision Tree with no depth limit was trained first to demonstrate overfitting:

Metric	Value	Meaning
Train RMSE	~0.0000	Tree memorized every training sample
Test RMSE	~0.7300	Failed to generalize to new data
Gap	~0.7300	Severe overfitting confirmed

An unconstrained tree splits nodes until each leaf contains a single training sample. The result is a model that memorizes rather than learns — useless in production.

2. Cross-Validation Results

A single train-test split can produce misleading results depending on which samples end up in each set. 5-Fold Cross-Validation addresses this by rotating the test set across 5 different portions of the data. The average RMSE across all 5 folds gives a stable, trustworthy estimate.

Model	CV RMSE (5-Fold)	Assessment
Linear Regression	0.7459	Consistent — no variance issues
Ridge Regression	0.7459	Identical to Linear — regularization had minimal impact
Decision Tree (Untuned)	0.9029	Worst performer — overfitting confirmed

3. Hyperparameter Tuning via GridSearchCV

GridSearchCV automates the process of finding optimal hyperparameters by testing every possible combination from a defined grid. Each combination is evaluated using 5-fold CV, ensuring the selected parameters generalize well rather than just fitting the training set.

Search Space:

- max_depth: [3, 5, 7, 10] — limits tree growth to prevent memorization
- min_samples_split: [2, 5, 10] — requires minimum samples before a split is allowed
- Total combinations tested: 12 | Total fits: 60 (12 x 5 folds)

Optimal Parameters Identified:

Parameter	Best Value	Effect
max_depth	10	Limits tree depth — reduces overfitting
min_samples_split	10	Prevents splits on small noisy groups
Best CV RMSE	0.6366	Best generalization score found

4. Model Comparison

Rank	Model	Test RMSE	R2 Score
1st	Tuned Decision Tree	0.6454	0.6821
2nd	Ridge Regression	0.7456	0.5758
3rd	Linear Regression	0.7456	0.5758

5. Final Model Justification

Selected Model: Tuned Decision Tree

Performance justification:

- Achieved the lowest Test RMSE (0.6454) — 13% better than Linear and Ridge (0.7456)
- R2 of 0.682 indicates the model accounts for 68.2% of price variation in the dataset
- California housing prices are influenced by non-linear factors (location, density, income thresholds) that linear models cannot model accurately

Overfitting reduction:

- Untuned tree gap: Train RMSE ~0.00 vs Test RMSE ~0.73 (gap = 0.73)
- Post-tuning: gap reduced significantly via max_depth=10 and min_samples_split=10
- CV RMSE (0.6366) closely aligns with Test RMSE (0.6454) — no signs of overfitting remain

Model trade-off analysis:

- Linear Regression: fastest, most interpretable — but assumes linear feature relationships
- Ridge Regression: adds L2 penalty to reduce variance — still limited to linear decision boundaries
- Tuned Decision Tree: captures complex patterns, controlled depth prevents overfitting — optimal for this dataset